Exercises Block 2

1. Introduction

In the exercises for Block 2 of the MSE module TSM ModAvi, you will apply again the theoretical concepts from the videos of Block 2 on physical properties of wave proagation as well as ground- and space-based aeronautical infrastructure. The exercises are divided into two parts: In Exercise 1 you will have to do coverage simulations of a terrestrial navigation system, in Exercise 2 you will have to do coverage analysis for space-based infrastucture.

2. Tasks

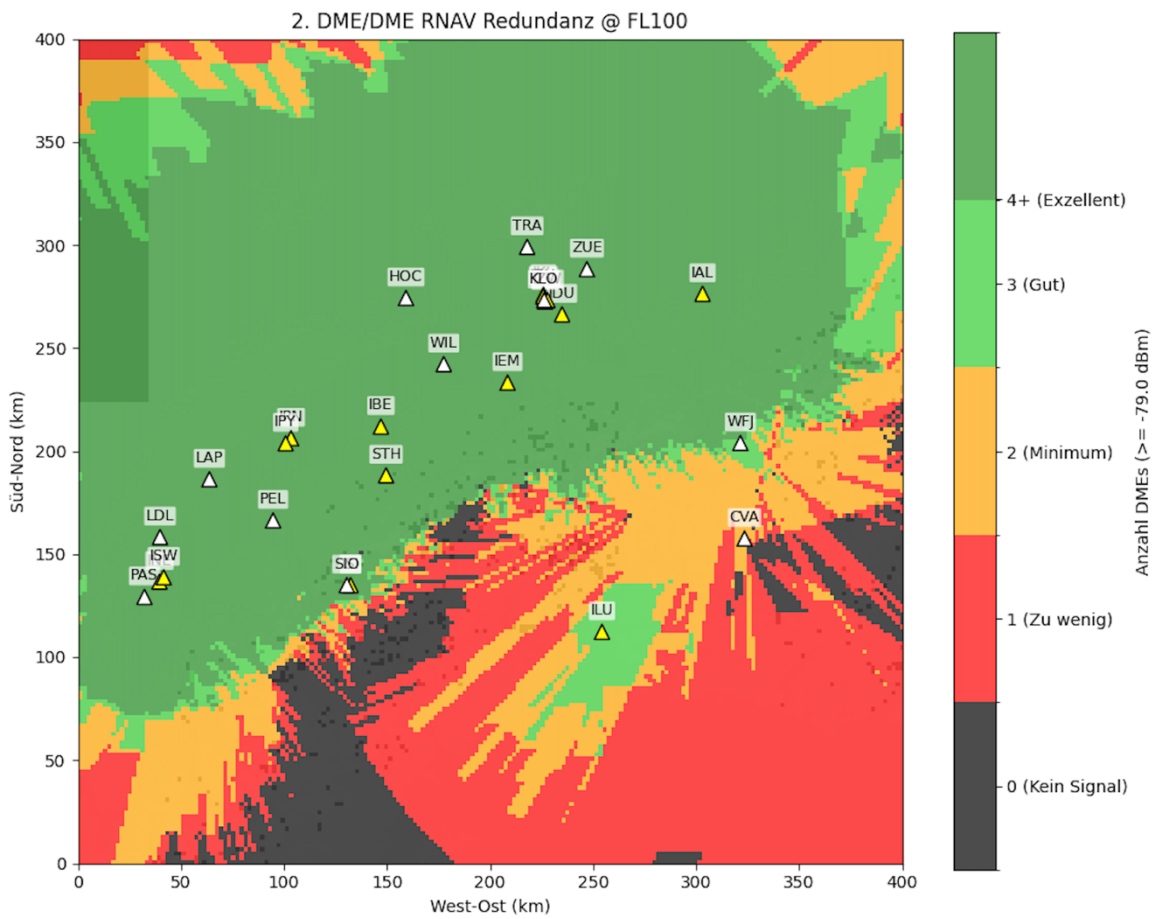
The overall objective of these exercises is to apply the concepts and principles taught in the videos of Block 2 in practical examples. Solve these tasks by writing, testing and analysing software code (in Python, Matlab or any coding language of your choice).

In general, go step by step, break the problem into smaller sub-problems, think about the interfaces, prepare a UML formulation of your code, define test cases and then implement using the help of LLMs.

Exercise 1:

Your task is to analyze the coverage of the existing network of DME stations in Switzerland at relevant altitudes taking into account their transmit power, physical propagation properties of the signal, as well as the terrain.

In preparation for this task please:

- https://www.ead.eurocontrol.int/ set up a free account, log in, go to SDO Reporting and download the list of all DMEs world-wide.
- Find the 27 DMEs in Switzerland
- For en-route DMEs use a transmit power of 1kW, for ILS/Terminal DMEs use a transmit power of 100W.
- Assume the minimum received signal power that is needed by an airborne receiver is -79dBm.
- Use SRTM data for terrain (or use the terrain tiff provided)

- Plot a map that shows the coverage of the DMEs, indicating how many DMEs are usable at that location. The output should look something like this: